



PROJECT-BASED LEARNING

PHASE-I EVALUATION

Project Title

HopeLine: Smart Hospital Resource and Blood Donor Matching System

Team ID: DSCPP-III-2026-T225

Department of Computer Science & Engineering

Graphic Era (Deemed to be University), Dehradun

Academic Session 2026-27

Team & Mentor Details



Team Information

Team ID: DSCPP-III-2026-T225

Semester: 3rd

Section: DS-1, DS-2, ML-4

Domain: *Healthcare Systems –
Data Structures & OOPS (C++)*

Team Members

1. *Shivansh Khanna – 2510012154*
2. *Piyush Srivastava – 2510370317*
3. *Parikshit Thakur – 2510380241*

Mentor

Mr. Sanjeev Kukreti

Interactions completed: 2

Problem Statement

During medical emergencies, hospitals often struggle to simultaneously find an available bed, ICU unit, or oxygen supply for an incoming patient, and to locate a compatible blood donor when a transfusion is urgently needed. Managing these resources manually or on a first-come-first-served basis causes delays, particularly when several critical patients arrive at once.

Motivation

- ▶ Bed and donor allocation are handled as separate, manual processes today
- ▶ Delays directly impact patient survival in critical cases
- ▶ Existing hospital software handles records, not real-time priority matching

Target Users / Stakeholders

- ▶ **Primary:** Hospital emergency staff and critical patients
- ▶ **Secondary:** Registered blood donors, hospital administration
- ▶ **Expected benefit:** Faster, fairer, automated emergency response

Project Objectives

1. *Build a C++ system modeling hospitals, patients, resources, and blood donors.*
2. *Implement priority-based bed/ICU/oxygen allocation and blood donor matching.*
3. *Integrate Data Structures (Hash Table, Priority Queue, Queue) with OOP concepts (classes, inheritance, polymorphism, STL).*
4. *Evaluate the system using sample data and edge cases (no bed/no donor available).*
5. *Demonstrate practical applicability for real hospital emergency triage.*

Key Objective: *Automatically identify the single best bed and the single best blood donor for each emergency patient in real time.*

Proposed Approach

1. *Patient / Donor Data Input*
2. *Priority Score Calculation*
3. *Hash Table + Priority Queue Matching Logic*
4. *In-Memory / STL-based Storage*
5. *Output: Best Bed / Best Donor*

Key Features

- ▶ Automatic priority scoring for patients
- ▶ Instant Hash Table lookup by type/blood group
- ▶ Priority Queue-based donor and bed ranking
- ▶ Overflow Queue for unmatched requests

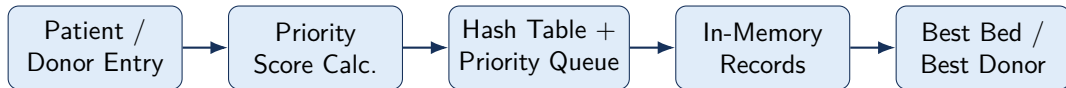
Instead of returning a plain list of options, the system computes a weighted matching score (severity, availability, distance, urgency) and returns the single best match – reducing decision time during emergencies.

Integration of PBL Course Concepts



Course	Concepts Applied	Project Module
Data Structures in C	<i>Hash Table, Priority Queue (Heap), Queue</i>	<i>Bed Allocation & Donor Matching Engine</i>
OOPs with C++	<i>Classes, Inheritance, Polymorphism, STL</i>	<i>Patient/Resource/Donor Class Design</i>

Requirement: Demonstrate meaningful integration of both subjects in **one** project.



Major Components

- ▶ Patient Module
- ▶ Resource Module (Bed / ICU / Oxygen)
- ▶ Donor Module

Data / Control Flow

- ▶ Patient entry triggers a priority score calculation
- ▶ Hash Table locates candidates; Priority Queue ranks them
- ▶ Hospital class returns the best bed and best donor

Technology Stack

- ▶ Programming: *C++17*
- ▶ Storage: *STL containers*
- ▶ Tools / Framework: *Standard C++, STL*
- ▶ IDE: *VS Code / Code::Blocks*
- ▶ Version Control: *Git / GitHub*

Development Methodology

1. Requirement Analysis
2. System Design
3. Implementation
4. Testing
5. Evaluation

C++ was chosen for direct control over data structures and OOP demonstration; STL containers keep the project focused on DS/OOP without extra database overhead.

Initial Progress & Team Contribution



Progress Achieved

- ▶ Background / literature study on triage systems
- ▶ Requirements identified
- ▶ Class structure & DS architecture prepared

Individual Contribution

Member	Role	Contribution
Shivansh	Lead	<i>Design, Priority Queue module</i>
Piyush	Developer	<i>Hash Table, donor matching</i>
Parikshit	Testing/Docs	<i>Edge-case testing, docs</i>

Roadmap, Expected Outcomes & References



Roadmap

1. Phase-I: Proposal & Design
2. Phase-II: Development
3. Phase-III: Final Implementation

Expected Outcomes

- ▶ Working C++ prototype / system
- ▶ Measurable lookup and ranking efficiency
- ▶ Practical triage applicability

References

1. *GeeksforGeeks – DS & Algorithms documentation*
2. *e-RaktKosh, Ministry of Health & Family Welfare*
3. *C++ Reference (cppreference.com)*
4. *Emergency Severity Index (ESI) triage guidelines*

Thank You
Questions & Suggestions